TABLE S6a

PAIRED REFERENCE AGREEMENT

| Clustering | Model | Means (%) | Means + fractions (%) | Paired change (pp) |
|---|---|---|---|---|
| Leiden | GPT-6 Astra | 52.86 | 58.12 | +5.26 |
| FlowSOM | GPT-6 Astra | 36.66 | 36.05 | -0.61 |
| SpatialSort | GPT-6 Astra | 36.13 | 32.38 | -3.75 |
| PIXIE | GPT-6 Astra | 41.32 | 35.64 | -5.68 |
| Leiden | Claude Opus 5.5 | 46.78 | 50.27 | +3.49 |
| FlowSOM | Claude Opus 5.5 | 37.48 | 36.82 | -0.66 |
| SpatialSort | Claude Opus 5.5 | 33.11 | 36.17 | +3.06 |
| PIXIE | Claude Opus 5.5 | 36.04 | 42.54 | +6.50 |
| Leiden | Gemini 4 Argon | 46.16 | 47.51 | +1.35 |
| FlowSOM | Gemini 4 Argon | 39.57 | 41.84 | +2.27 |
| SpatialSort | Gemini 4 Argon | 33.31 | 33.80 | +0.49 |
| PIXIE | Gemini 4 Argon | 38.70 | 36.40 | -2.30 |
| Leiden | DeepSeek V4.1 Flash 1017 | 42.58 | 44.96 | +2.38 |
| FlowSOM | DeepSeek V4.1 Flash 1017 | 28.38 | 27.50 | -0.88 |
| SpatialSort | DeepSeek V4.1 Flash 1017 | 22.62 | 23.06 | +0.45 |
| PIXIE | DeepSeek V4.1 Flash 1017 | 35.31 | 26.56 | -8.75 |